

Education

Duke University: *B.S.E. Mechanical Engineering, Aerospace Engineering Certificate; 3.97 GPA* Sep 2024 – May 2028

Courses: Space Systems Design, Aircraft Propulsion, Aeroelasticity, Heat & Mass Transfer, Fluid Dynamics, Control System.

Experience & Leadership

Propulsion Project Lead - *Duke AERO Rocketry Society* Jun 2026 - Present

- Chief Engineer for Project Prometheus: Nitrous Oxide - IPA pump-fed rocket engine targeting 3500 N thrust and 14,000 ft.
- Architect high-level system design, sizing & thermal analysis of regeneratively-cooled pintle injector combustion chamber.
- Own clean-sheet design, analysis, manufacturing and testing of the electric centrifugal propellant pump.
- Dictate full-scale fluid system architecture, including flight tanks, fill & QD system, active pressurization & engine controller.

Propulsion Engineer, Avionics & Controls Lead Engineer - *Duke AERO* Jun 2025 - Jun 2026

- Led full-cycle design, analysis, and manufacturing of tri-blade airbrake system for major 40% weight reduction.
- Owned full-cycle redesign of canards system for miniaturization & 20% reduction in manufacturing & integration time.
- Developed Monte Carlo framework for control algorithm characterization (gain-scheduling, PID, adaptive MPC) for precise apogee targeting of rocket to 30,000 ft with airbrakes and canards.
- Spearheaded novel composite designs techniques: carbon-fiber overmolded control surfaces & forged fiberglass hatch.
- Led end-to-end manufacturing of a 6.125" OD x 7.5" aluminum control-stack housing, owning machining strategy, setup, programming, and optimization to reduce cycle time by 50%.
- Full-cycle ownership of design, manufacturing & characterization of new propellant manifold & impinging injector design.

Avionics, Structures, Propulsion Design Engineer - *Duke AERO* Sep 2024 - Jun 2025

- Led **FINSight V2**: novel dual-layer PCB with embedded accelerometers & DAQ system to characterize rocket fin flutter.
- In charge of design & analysis of IP-67 rated, blast-proof **black box** housing & PCB, validated with ANSYS FEA.
- Led research on **canard**-tail fin aerodynamic coupling & flutter for roll-stabilization. Developed closed-loop PI ramping control process. Selected for podium presentation at IREC 2025 as an outstanding project.
- Conducted NOX & Kerosene injector simulations on ANSYS Fluent to validate injection pattern and pressure losses.
- Manufactured flight-critical **propulsion hardware**, including 6" solid rocket motor closures, nozzle & APCP propellant grains.

Employment

R&D Engineer Intern - *Keshavarz Fluid Dynamics Group* May 2026 - Present

- High-speed photography of hydrodynamic instabilities in rotary atomization. Fuel injector design & combustion processes.

Design Engineer - *Duke DesignHub* Dec 2025 - Present

- Provide engineering consultation & product design services for Duke researchers & hospital clinicians.
- Designed & fabricated a LED-integrated dual-camera surgical headgear with articulated stay-put arms and cooling fins.

Machinist & Teaching Assistant - *Duke Pratt Student Machine Shop* Sep 2025 - Present

- Support machining classwork & design projects by providing machining expertise and shop safety oversight.

Mechanical Engineering Intern - *GEA Group Vietnam - Liquid & Powder Technology* Apr 2024 - Aug 2024

- Designed & documented P&IDs; conducted on-site tank, piping & valves inspections & maintenance for F&B factories.

Robotics Research Intern - *Chirikjian Lab, National University of Singapore* Mar 2023 - Jan 2024

- 7-DOF robot arm pick & pour simulations with MATLAB, ROS-2 & PyBullet.

Publications

Predicting The Evolution Of Rocket Engines: An Analysis Of Past Trends And Prediction Of Future Trends

Upcoming issue of the *Philosophical Transactions of the Royal Society A*

Technologies

Skills: ANSYS Mechanical & Fluent Turbo, GD&T, Tolerance Stackup, Combustion & Propulsion Systems, Composite Layup & Molding, Tooling Design, DFM, CAD/CAM (OnShape, SolidWorks, AutoCAD, MasterCAM, Siemens NX), Aircraft Design Suite (AVL, XFOIL & XFLR5, OpenVSP, MachUp), PCB design (Altium, Eagle)

Programming & Control: Python, C, C++, Java, FORTRAN, MATLAB Simulink, MIPS Assembly, ROS, LABView & NI-DAQmx